

Research paper

Costs relations in the hydrocarbons supply chain project

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ABSTRACT

Most of the hydrocarbons supply chains are carried out using marine and pipeline transports. The use of tankers or pipelines on particular routes is determined by economic reasons. The problem of cost estimation in the hydrocarbons supply chains has not been covered in scientific publications. The purpose of this paper is to develop a method for analyzing the cost relations in the oil supply chain of an energy project in the Eastern Siberia Pacific Ocean (ESPO) Region. The results obtained using the developed method showed that the total cost of transporting one ton of oil for 100 km by using ESPO pipelines are 6–8 times greater than the cost for a similar maritime shipping. It also shows that the cost of oil transportation will significantly increase if the pipeline capacity utilization decreases and the interest rate increases.

1. Introduction

Depending on the structure of the supply chain, supply chain costs can account for anywhere from a few percentage points to several tens of percent of the cost of supplied hydrocarbons. To understand the factors contributing to the cost of oil on world markets, it is crucial to research the expenses associated with transporting hydrocarbons in supply chain. Therefore, studying the transportation costs of hydrocarbons is an important objective. The existing transportation methods yield to the pipeline and maritime transportation in the price competition when oil is delivered over long distances. About 30 % of international transportation in the world is accounted for by oil multimodal transportation, which includes both pipelines and sea transport. This phenomenon is one of the significant concerns in the global supply chains.

The Eastern Siberia-Pacific Ocean (ESPO) oil delivery project is used as a case study. The ESPO project consists of pipeline and maritime components. The ESPO pipeline system consists of two parts (ESPO-1 and ESPO-2) and has a total length of 4800 km). The projected capacity of ESPO-1 is planned to be 80 million tons per year. Once it reaches Skovorodino,¹ about 30 million tons of oil are flown through ESPO-1 and sent to an adjacent pipeline from Skovorodino to Daqing, China. The remaining 50 million tons of crude oil per year have to flow through

Skovorodino to the second segment of the pipeline system (ESPO-2) to the port of Kozmino.² The "Transneft" company is the owner and the operator of the ESPO oil pipelines system. The pipeline from the oil fields of Siberia and the specialized Kozmino port is an integral element of the ESPO pipeline system. Therefore, the costs of the port operations are not separated from the pipeline system costs. In the ESPO delivery project, oil is delivered through the pipeline to the transition port Kozmino and further by oil tankers to the customer's terminals. The ESPO project is the largest multimodal project for the supply of Siberian crude oil to China and demonstrates the cost ratio between sea and pipeline deliveries.

The multimodal project's efficiency depends on costs related to each transport mode involved in transportation. To elaborate on the efficient supply chains and to expand the geography of oil transportation using these two transportation means, we should determine which of them is the most advantageous in terms of transportation costs. The selection of appropriate cargo delivery arrangements may produce significant savings for the majority of the supply chain projects.

The objective of this paper is to estimate cost relations in oil transportation by sea and by pipeline. In order to achieve this objective, the following tasks should be resolved. (i) We develop a method to calculate the costs for hydrocarbons transportation by sea and by pipeline and determine the influence of the main economic factors on the total costs

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E-mail address: vklepikov@hse.ru (V. Klepikov).¹ Skovorodino is an administrative^[1] and municipal^[7] district (raion), one of the twenty in Amur Oblast, Russia.² Kozmino is an oil port 85 kilometres (53 miles) southeast of Vladivostok on the Nakhodka Bay near Nakhodka, Primorsky Krai, Russia and close to Russia's borders with China and North Korea, on the coast of the Sea of Japan.

Nomenclature			
ESPO	Eastern Siberia-Pacific Ocean pipeline system consists of two parts	R	tanker fuel consumption while moving (ton per day)
ESPO-1	first part of ESPO pipeline system	p_f	value of the annual increase in the cost of fuel
ESPO-2	second part of ESPO pipeline system	m	number of cost components in formula (5)
$W_1(t)$	annual value of depreciation costs	W_C	average value of expenditures per ton of transported cargo for the whole duration of the project
T	duration of project in years	$M(t)$	total weight of the goods delivered per year by the transport unit (pipeline or tanker)
C_u	value of the initial investment in the project	P_1	average cost of the capital for the whole duration of the project
$W_2(t)$	annual capital loan expenses	P_2	average operating cost for the whole duration of the project
t	current year of the project	P_3	fuel cost for tanker for the whole duration of the project
j	interest rate	W_T	total costs value for the project duration (for tankers)
$W_3(t)$	insurance cost	$N_l(t)$	annual number of tankers
a	percentage for the insurance services	l	tanker type
p_s	rising cost of insurance per year	G_l	tankers' cargo capacity
$W_4(t)$	operating costs	$V(t)$	annual volume of oil transportation
E	annual operating costs	$W_C^{100km}(k)$	cost of transportation of one ton of goods over a distance of 100 km
p_h	annual increase of operating costs	L_1	length of pipeline ESPO-1
W_T	total cost value for the pipeline for the period of operations	L_2	length of pipeline ESPO-2
$K(t)$	discount coefficient	L_3	distance of sea shipping
d	discount rate	Z	utilized capacity
$W_5(t)$	cost of fuel per year for one tanker		
T_v	time of the tanker actual movement per year		
F	cost of fuel (US\$ per ton)		

of oil transportation. (ii) We compare the costs for oil transportation by sea and by pipeline for the existing project.

The major finding of this research reveals that the costs of transporting the same amounts of oil by sea tankers are well below the costs of oil transportation by pipeline. This finding demonstrates a significant efficiency of transportation by sea when compared to the pipeline. The results obtained demonstrate the effectiveness of the methodology that is used. The proposed method for modeling and comparing costs in the supply chain elements can be used to study various oil supply options when developing tariffs for new ones and estimating costs of existing ones for multimodal oil supply projects. The contribution to the literature is that this study fills a gap in the publication on cost relations in oil supply chains. This research will allow the development of a promising direction in energy logistics.

The remainder of the paper is organized as follows. Section 2 provides a review of the literature. Section 3 presents the methodology and Section 4 the results. Finally, Section 5 discusses the results and Section 6 concludes.

2. Review of the literature

Cost-related problems in the delivery of goods by various modes of transportation have been considered by a number of authors. Issues with the main studies in the field of oil logistics as well as the suggested project have prompted the current authors to conduct this study. In (Alizadeh and Nomikos, 2004), the possible connections between futures, spot oil prices and tanker freight rates were analyzed. However, a clear connection between these markets was not established. Alizadeh and Talley (2011) investigate the factors influencing tanker freight rates and shipping times, and analyze tanker shipping contracts from 2006 to 2009. The results of the study show that the length of the lead time is an important factor in determining the ocean freight rate. The age of the vessel, the type of hull of the vessel, and voyage routes also have a slight influence on the freight rate.

Adland and Cullinane (2006) analyze oil freight rate fluctuations using a Markov’s model. They demonstrate that the dynamics of the spot freight rate in the oil transportation market can best be described by a non-linear stochastic model. They also reveal that the spot freight rate is

mean reverting only in the extremes of the empirical range, and that the volatility of the freight rate changes increases with the level of the freight rate. Adland et al., (2016)) developed a model for setting up freight rates for Capesize and VLCC tankers. The author show that, although market conditions and routes remain the most influential covariates, the characteristics of charterers, owners and of their matches are also significant microeconomic determinants of the freight rate level. The model, which establishes the connection between market rates on dry cargo ships and oil tankers, was presented in article (Tsouknidis, 2016). The process of tanker chartering optimization for China oil refineries was developed by Wang et al. (2013) using mathematical models. It was assumed that freight rates comply with the Brownian motion and the arrival of tankers is subject to the Poisson law. Then, the process can be narrowed down to a differential equation in partial derivatives. Actual examples have confirmed the advantage of this method.

Modes of oil transportation by sea are studied in Lun et al. (2013). The authors examine the industry, depict the shipping market and its structure and forecast the location plan and the transportation chain. They conclude that oil transport management provides a concise yet extensive coverage of the oil transport industry’s history and a guide for its future development. A programming model to determine maritime transportation routes for crude oil, taking both connectivity reliability and transportation costs into consideration, was developed in Wang et al. (2023). Those authors proposed a method which captures the structure of risk events. Panayides et al. (2011) developed two models to evaluate the effectiveness of shipping companies. The authors used several dry cargo shipping, oil product shipping and container shipping companies. They show that using the tanker companies is the most efficient, while using the dry cargo companies is the least efficient in shipping. Using the structural vector autoregressive (SVAR) approach, Aizemberg et al. (2014) propose operational models for oil transport by tankers. A feasible solution is constructed for the original problem using the obtained columns. Shi et al. (2013) investigate the correlation between oil price fluctuations and the tanker freight market. Agra et al. (2013) address the issue of oil delivery by tanker fleets for short distances between islands. The oil inventory is regulated by market demand, and ports have a limited time for ship loading and unloading.

Therefore, tankers spend a long time waiting in queue for unloading. In the article, a mathematical model was proposed to address this issue. [Fei et al. \(2020\)](#) determined the fluctuation characteristics of the offshore crude oil transportation system and the impact of critical events on them. Using the model presented, numerical results were obtained that confirmed the effectiveness of the method used for the three transport lines under consideration.

Specific features of pipelines in Russia are discussed in ([Pototskaya et al., 2016](#)). Those authors discuss the research on transformation of territorial organization of Russian pipeline transport in the post-Soviet time, taking into account its relations with neighboring countries. The study identifies general ways of such transformation which includes the influence of Russia's desire to escape from the dictate of the transit countries exporting energy, the impact of Russia's strive for the transportation of gas and oil extracted in the Caspian Sea basin, the influence of the strive for the transportation of oil and gas in the Asia-Pacific region. They show the correlations in the development of pipeline transport in the post-Soviet period with the development of infrastructure of the country's sea transport. [Vatansever \(2017\)](#) examines the entire network of oil and gas exports from Russia, identifying a significant excess of pipeline capacity and defining a pipeline development concept. The foreign energy policy of Russia has been studied, and three explanations are given for Russia's excess oil and gas export capacity. [Balat \(2006\)](#) analyzes the features of creating the Baku-Tbilisi-Ceyhan oil export pipeline. The advantages of constructing such a large oil pipeline in the Caspian oil-rich region are shown by this author, along with the capacity estimates and the budget for pipeline construction. Economic data for the construction and utilization of 412 pipelines are presented in [Rui et al. \(2011\)](#). The construction cost of the pipeline depends on its diameter, length, throughput capacity, time of construction and geographical location. The issues related to the pipeline networks design were analyzed in [André et al. \(2013\)](#). This study presents a new methodology for the simultaneous determination of the topology and of the diameters of hydrogen transport networks. These two problems were generally solved separately. For instance, the determination of the optimal topology is generally based on some basic criteria such as the minimum length of the network. It shows that the two stage-approach generally used (first looking for a network topology of minimal length and then optimizing the diameters for this fixed topology) is not sufficient. On the opposite, their case studies indicate that increasing the total length of the network can help decrease the network cost by using smaller diameters for some pipes.

A single model is proposed in [Dey \(2004\)](#) for estimation and selection of pipelines building projects. The market analysis determines pipelines throughput and supply and demand points. Subsequent, technical analysis identifies technological options and economic and financial analysis then derives the least cost option among all technically feasible options. The subsequent impact assessment tries to justify the selected option by addressing environmental and social issues. Based on the findings of the West to East Oil Transmission project, [Lin et al. \(2024\)](#) use a time-varying difference method to study the impact of oil transportation strategy on the performance of enterprises. The results show that the project significantly improves the production performance of enterprises along the pipeline and enhances the operational performance of the other enterprises in neighboring cities. To reduce energy consumption, [Liu et al. \(2022\)](#) investigated an existing oil pipeline with a view to ensuring safe production. The developed mathematical model of oil pipeline energy consumption uses an artificial bee colony algorithm. In order to optimize energy consumption and ensure the efficient operation of the oil pipeline, the frequency of the pump speed was adjusted.

The intermodal transportation model, which consists of marine, rail and road modes of transport to deliver cargoes, was developed in [Klepikov and Klepikova \(2023a\)](#). The influence of vehicle loading on the costs of delivery is shown, and a comparison of the cost components for water, motor and railway transport was carried out. The authors

conclude the cost of air transport is greater than other methods of delivery and the least expensive is the transportation of water. Motor transportation had some advantages over railway for short distances, but when the length of the route increased, railway delivery was more profitable. The cause is the carrying capacity of the railway trains. Oil deliveries using rail and pipeline transport are considered in [Klepikov and Klepikova \(2023b\)](#). The authors demonstrate that the annual growth rate of cumulative emissions from storage of oil in refinery tank farms and total emissions in the entire supply chain (compared to 2009) are consistent and show the highest values, compared to other elements of the supply chains during the sample period. Lower values are also demonstrated by the growth of cumulative emissions from pipeline delivery. The rate of change in emissions from rail delivery is inferior to the annual change in cumulative emissions (except for 2010) for other elements of the supply chain and decreases slightly in 2011–2015.

[Zhang et al. \(2013\)](#) establishes a framework for assessing the security of oil imports from the perspective of the supply chain establishment process and constructed a two-stage model to assess China's oil supplies. The results indicate that the risks associated with China's oil imports continue to increase at every stage of the supply chain. Among these risks, the threat of external dependence has emerged as the most significant challenge. [Kazemi and Szmerekovsky \(2015\)](#) use a linear programming method to develop a model for determining the optimal oil supply for oil refineries. The model minimizes the cost of the oil supply chains, refineries and centers of product distribution. That study was based on the US oil industry and the US transportation network data. The authors conclude that the optimal supply chain design is diverse when planning for a single mode and multiple mode transportation. That is, it is crucial to consider multimodal transportation when locating the facilities at the strategic level. Failing to do so will result in additional costs and a sub-optimal supply chain configuration. Thus, their proposed multimode model shows a more cost-efficient structure by optimizing the location of the distribution centers along with selecting the best transportation mode or modes at any point in the supply chain to meet the total demand.

[Oliveira et al. \(2016\)](#) analyze the system from marine terminal to crude oil pipeline to oil refinery and developed an optimization model of its function, given the conditions of uncertainty concerning oil supplies to the terminal. A solution was proposed on how to select an optimal pipeline diameter with maximum cost efficiency. To minimize overall costs for oil refineries, [Guajardo et al. \(2013\)](#) provide a solution to address an integrated planning of production, transportation and sales. While the total contribution is higher in the integrated approach, it has also been found that the sellers' contribution can be considerably lower. Therefore, the authors suggest contribution sharing rules to achieve a solution where both the company and the sellers attain a better outcome under the integrated planning.

[Shahnazi et al. \(2023\)](#) analyzed the global structure of oil trade. They studied the stability of the global trade network and the stability of countries in the export and import of oil. An index was developed to assess the effectiveness of each country in oil trading networks, and a sustainability measure was introduced to ensure the stability of countries in the global structure of oil trade. [Bilgin and Ellwanger \(2024\)](#) represent a model of the world oil market that uses information about world oil consumption. Under the assumptions made, data on the global fuel consumption provides a relatively accurate picture of the structural parameters of the global oil market. [Al-Othman et al. \(2008\)](#), propose a stochastic model for supply chain planning for oil companies. Using this model, those authors show that economic uncertainty encourages oil-producing countries to maintain a balance between the processing and export of oil. In [Aboudi et al. \(1989\)](#), the main stages of the research project for oil fields development was analyzed as well as a related transportation systems design. The authors developed a mathematical model, determined the solution methods and described ways to implement them. The developers of the project use proposed algorithms. [Toppur and Sanyal \(2020\)](#) simulated oil supplies by tankers and

pipelines to refineries in India from the Persian Gulf, Africa, and Asia. The scaled-up problem has been modelled, as a transshipment network. Then the concrete instance of the problem was solved using an LP solver, which was followed by post-optimality analysis of the solution. The flow values on all arcs, and optimal product mix corroborate actual decisions. Precise shipping requirements obtained from the solution are also shared in advance with marine transporters to advance supply chain coordination.

This paper will analyze the cost structure in the oil supply chain, which has not been done in the literature, and thus our findings will also be new.

3. Methodology

In the project, which was the basis for this research, oil is delivered through a pipeline to the transition port and further by tankers to the customer's terminals. This section solves the problem of constructing mathematical relationships for calculating the costs of pipeline and maritime transport. Taking into account the infrastructure and the vehicles, the total costs of transportation in this project consist of the following main components: the costs of capital, operating expenses and the cost of fuel.

3.1. The costs of capital

If a linear method of accounting is used, the annual value of depreciation costs is determined by the following ratio:

$W_C^{100km}(k)W_1 = (C_u + C_{moder} + C_{repair} - C_{liquidation})/T$ where C_u is the value of the initial investment in the project, C_{moder} is investments for modernization, C_{repair} is repair expenses throughout the project, $C_{liquidation}$ is liquidation cost of fixed assets, T - duration of project in years.

The most significant component in the expression for W_1 is C_u . Other components can be neglected. Therefore, the cost of depreciation is represented by the expression:

$$W_1(t) = C_u/T \quad (1)$$

Annual capital loan expenses W_2 can be determined using the following ratio:

$$W_2(t) = C_u \cdot \left(1 - \frac{t-1}{T}\right) \cdot j \quad (2)$$

where t is current year of the project and j is interest rate.

3.2. Operating expenses

A significant portion of the operating expenses is attributed to the insurance cost. Taking into account the annual decrease of the insurance value and the increase in the cost of insurance services, the insurance cost can be determined by the following equation:

$$W_3(t) = a \cdot C_u \cdot \left(1 - \frac{t}{T}\right) \cdot (1 + p_s)^t \quad (3)$$

where a is the percentage for the insurance services and p_s is the rising cost of insurance per year.

Another part of the operating expenses (operating costs) is administration, supply and stocks, maintenance and repairs, salaries, etc. These costs can be described by the following expression:

$$W_4(t) = E \cdot (1 + p_h)^t \quad (4)$$

where E is the initial value of the annual operating costs and p_h is the value of the annual increase of operating costs.

Summarizing the above, the total cost value W_T for the pipeline for the period T of operations can be determined by the ratio:

$$W_T = \sum_{t=1}^T \sum_{i=1}^m W_i(t) \cdot K(t) \quad (5)$$

where $K(t)$ is discount coefficient and $K(t) = (1 + d)^{-t}$, and d is the discount rate.

3.3. Fuel costs

The cost of fuel per year for one tanker can be calculated by the following formula:

$$W_5(t) = T_v \cdot R \cdot F \cdot (1 + p_f)^t \quad (6)$$

where T_v is the time of the tanker actual movement per year, F is the cost of fuel (US\$ per ton), R is tanker fuel consumption while moving (ton per day) and p_f is the value of the annual increase in the cost of fuel.

3.4. Other expenses

In addition to the above, there can be additional expenses. However, the value of these extra expenses usually does not exceed a small percentage of the total costs of vehicle operations. Therefore, these extra costs can be neglected.

3.5. Total costs

The number of cost components in formula (5) is $m = 4$ for pipelines, and $m = 5$ for ships. To calculate costs per ton of transported cargo, it is possible to express $W_T(5)$ in terms of W_C , $M(t)$ and $K(t)$ using the following equation:

$$W_T = \sum_{t=1}^T W_C \cdot M(t) \cdot K(t)$$

where W_C is average value of expenditures per ton of transported cargo for the whole duration of the project and $M(t)$ is total weight of the goods delivered per year by the transport unit (pipeline or tanker). Therefore, W_C can be determined by using the Eq. (5):

$$W_C = \frac{\sum_{t=1}^T \sum_{i=1}^m W_i(t) \cdot K(t)}{\sum_{t=1}^T M(t) \cdot K(t)}$$

We can decompose this equation into the following components: cost of the capital (P_1), operating costs (P_2) and fuel cost (P_3 - for tanker)

$$P_1 = \frac{\sum_{t=1}^T \sum_{i=1}^2 W_i(t) \cdot K(t)}{\sum_{t=1}^T M(t) \cdot K(t)}, P_2 = \frac{\sum_{t=1}^T \sum_{i=3}^4 W_i(t) \cdot K(t)}{\sum_{t=1}^T M(t) \cdot K(t)}, P_3 = \frac{\sum_{t=1}^T W_5(t) \cdot K(t)}{\sum_{t=1}^T M(t) \cdot K(t)}$$

and can be represented as follows:

$$W_C = \sum_{q=1}^n P_q, \text{ for the pipeline, for tankers} \quad (7)$$

The value W_C is the key value while setting up tariffs for the considered modes of transport.

For tankers, W_C can be determined using Eq. (1) - (7). For pipelines, W_C can be determined using Eqs. (1) - (5), and (7).

For tankers, total costs value W_T for the project duration T can be determined by Eqs. (1) - (6) with ($m = 5$) by the equation:

$$W_T = \sum_{t=1}^T \sum_{i=1}^m W_i(t) \cdot K(t) \cdot N_i(t) \quad (8)$$

where $N_i(t)$ is the annual number of tankers and i is tanker type.

3.6. ESPO project features

Ninety percent of the oil shipped (ESPO) from the Kozmino port goes to the ports of China, Japan and South Korea. The distances from Kozmino to Dzhoushan-Ningbo, Dalian, Chiba and the Ulsan port are 1871 km, 1898 km, 1647 km and 891 km, respectively. Therefore, the average estimated distance of sea shipping is ~ 1600 km.

Taking into consideration the average tanker speed, a one-way trip is approximately 3 days (6 days round trip). The estimated time to load or unload the tanker is 2 days. In total, the full trip of the vessel (including loading and/ unloading) is approximately 10 days, which amounts to 36 trips per year. Thus, the number of days per year when the ship is actually on sea is defined as 6 days $\times$ 36 = 216 days. For further calculations, it is assumed that only one type of the following tankers, Panamax, Aframax or Suezmax, is used for transportation.

To determine the appropriate number of tanker tonnage required for the transport of a given volume of oil per year, the following ratio is used:

$$N_l(t) = \frac{V(t)}{G_l \cdot 36}, l = 1 - 3 \quad (9)$$

where $N_l(t)$ is the annual number of tankers, G_l is tankers' cargo capacity for Panamax ($l = 1$), Aframax ($l = 2$) and Suezmax ($l = 3$) and $V(t)$ - annual volume of oil transportation.

The fractional part of $N_l(t)$ means that one of the tankers operated for less than one full year. In this example $N_l(t)$ and $V(t)$ are constants. In our example, the annual cargo traffic is constant; therefore, the number of involved oil tankers is also a constant number. One commonly used cost characteristic is the cost of transportation of one ton of goods over a distance of 100 km ($W_C^{100km}(k)$). In our case, the total length of a multimodal delivery consists of a length of pipeline ESPO-1 (L_1), ESPO-2 (L_2) and the distance of sea shipping (L_3).

To calculate the cost per 100 km for the abovementioned components of the delivery, the total value of the cost for each given segment (ESPO-1, 2 or sea transportation) is divided by its length and multiplied by 100.

$$W_C^{100km}(k) = \left(\frac{W_C}{L_k} \right) \cdot 100, k = 1 - 3 \quad (10)$$

4. Results

The initial data for all calculations are presented in Table 1. If different parameters are used, they are specifically marked. The

parameter values (#2, 4, 5, 6, 9, 10, 14, 15, 16) indicated in Table 1 represent the average values for pipelines and tankers involved in the transportation of oil under the ESPO project (Transneft annual reports, 2022) for the period of its activity from 2013 to 2022.

Route of Siberian oil delivery starts from pipeline oil transportation, that is why, it is necessary to consider the costs occurring for ESPO project.

These diagrams (Fig. 1 and Fig. 2) represent the costs to transport one ton of oil through the pipelines for the distance corresponding to the total length of these pipelines at full pipelines capacity ($Z = 100\%$). P_1 (P_1 on Figs. 1,2) is the cost of capital, P_2 (P_2 on Figs. 1,2) is the operating costs.

Fig. 1 shows cost components of transportation of one ton when changing the interest rate for ESPO-1.

Fig. 1 indicates that total costs ($W_C = P_1 + P_2$) for ESPO-1 grow in linear fashion as the interest rate increases. The value of the operating costs ($P_2 = \$20.992$) remains unchanged, and the value of the cost of capital (P_1) changes when the value of the interest rate (j) changes as follows: $j=0$, $P_1 = \$3.175$; $j=2\%$, $P_1 = \$5.474$; $j=4\%$, $P_1 = \$7.773$; $j=6\%$, $P_1 = \$10.072$. The total costs (W_C) with a zero interest rate ($j = 0\%$) is \$ 24.167 and with an interest rate of 6 % ($j = 6\%$) is \$ 31.063. The indicated changes in values are determined in accordance with Eq. (1)-(4) and (7). Since the interest rate (j) does not affect the value calculated using Eq. (1), but acts as a linear factor in Eq. (2), it linearly influences the value of P_1 . Conversely, the interest rate does not impact the values derived from Eqs. (3) and (4), thus not affecting the value of P_2 . In this scenario, the costs associated with P_1 and P_2 are elements used in

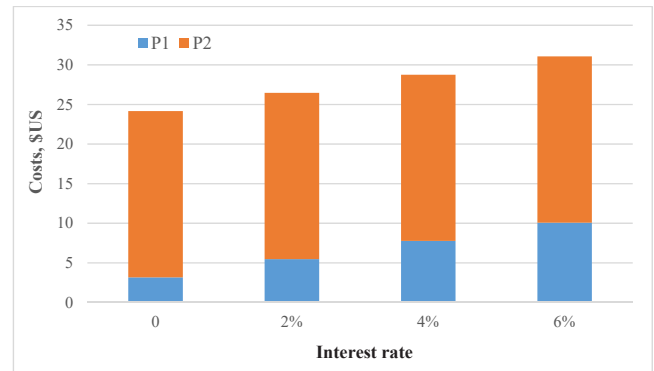


Fig. 1. Cost components for transportation of one ton through ESPO-1.

Table 1
Initial data.

#	Parameter	Symbol	Unit	Panamax	Aframax	Suezmax	ESPO-1	ESPO-2
1	Number of years of the project	T	year	20	20	20	50	50
2	Discount rate	d	%	3	3	3	6	6
3	*Interest rate	j	%	2	2	2	4	4
4	Rising cost of insurance per year	p_s	%	4	4	4	1	1
5	Share of the costs of insurance	a	%	1.7	1.7	1.7	0.5	0.5
6	Cargo capacity	G_l	1000 ton	70	110	160		
7	*Utilized capacity	Z	%	50	50	50	100	100
8	Distance of the route	L_k	1000 km	1.6	1.6	1.6	2.7	2.1
9	Rate of fuel consumption	R	ton/day	50	100	125	-	-
10	Fuel costs	F	\$/ton	320	320	320	-	-
11	Oil tanker running time for the year	T_v	day	216	216	216	-	-
12	Size of investments	C_u	mil \$	40	45	55	12 700	8 100
13	Annual volume of transportation	$V(t)$	mil ton	50	50	50	80	50
14	Annual operating costs	E	mil \$	2.727	2.760	3.088	648	315
15	Rising of operating costs per year	p_h	%	3	3	3	5	5
16	Rising of fuel costs per year	p_f	%	5	5	5	-	-

Source: Transneft annual reports (2022).

* parameters that can be modified during the computation.

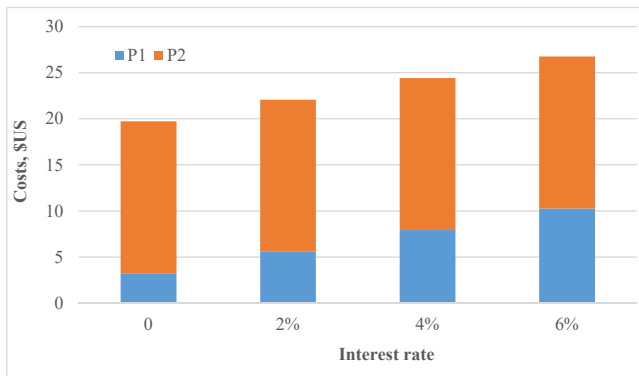


Fig. 2. Cost components for transportation of one ton through ESPO-2.

calculating the total costs ($W_C = P_1 + P_2$) for ESPO-1. Consequently, one of the terms (P_1) in the total costs calculation ($W_C = P_1 + P_2$) varies linearly based on the value of j , while the other term (P_2) remains constant.

Fig. 2 shows cost components of transportation of one ton when changing the interest rate for ESPO-2.

Fig. 2 indicates that total costs $W_C = P_1 + P_2$ for ESPO-2 grow in linear fashion as the interest rate increases. The value of the operating costs ($P_2 = \$16.478$) remains unchanged, and the value of the cost of capital (P_1) changes when the value of the interest rate (j) changes as follows: $j=0$, $P_1 = \$3.240$; $j=2\%$, $P_1 = \$5.586$; $j=4\%$, $P_1 = \$7.932$; $j=6\%$, $P_1 = \$10.278$. The total costs (W_C) with a zero interest rate ($j = 0\%$) is \$ 19.718 and with an interest rate of 6 % ($j = 6\%$) is \$ 26.756. The dependence is explained by the values obtained using Eq. (1)–(4) and (7). The interest rate (j) linearly affects the value determined by Eq. (2), which, in conjunction with the value determined by Eq. (1) (independent of j), are elements in determining P_1 . Therefore, the value of P_1 changes linearly with a change in j . The interest rate does not impact the values derived from Eqs. (3) and (4) that constitute the value of P_2 . Therefore, j does not affect the value of P_2 , and it remains constant when j changes. The total costs ($W_C = P_1 + P_2$) for ESPO-2 are determined by the sum of the values P_1 and P_2 , where P_1 changes linearly, while P_2 remains constant as j changes.

Diagrams in Fig. 3 and Fig. 4 are created using Eqs. (1)–(5) and (7). These diagrams represent a correlation between the cost of transportation of one ton of oil and the pipeline capacity utilization ($Z = 20\% - 100\%$). The minimum interest rate ($j = 0\%$) was used for these calculations. P_1 (P1 on Figs. 3,4) is the cost of capital, P_2 (P2 on Figs. 3,4) is the operating costs.

Cost components for the pumping of one ton of oil (W_C) for the ESPO-1 are shown in Fig. 3.

Fig. 3 demonstrates that total costs ($W_C = P_1 + P_2$) of the

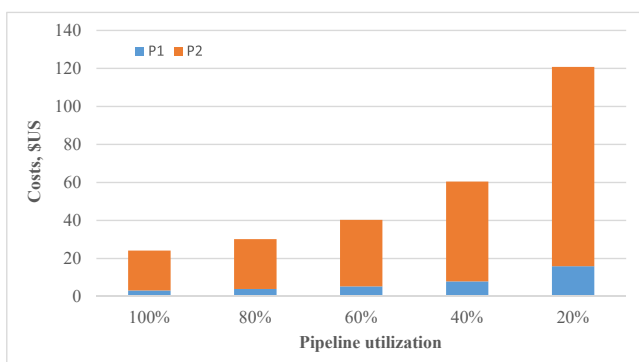


Fig. 3. Cost components for delivery of one ton when changing the capacity of ESPO-1.

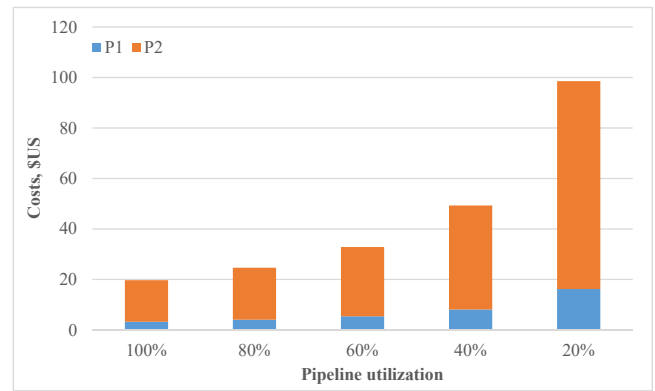


Fig. 4. Cost components for delivery of one ton when changing the capacity of ESPO-2.

transportation of one ton of oil for ESPO-1 increase significantly along with a decrease in the pipeline capacity utilization. When the pipeline capacity utilization $Z = 100\%$ (the cost of capital) $P_1 = \$3.175$ and (the operating costs) $P_2 = \$20.992$; when $Z = 80\%$, $P_1 = \$3.969$, $P_2 = \$26.240$; $Z = 60\%$, $P_1 = \$5.292$, $P_2 = \$34.986$; $Z = 40\%$, $P_1 = \$7.938$, $P_2 = \$52.479$; when $Z = 20\%$, $P_1 = \$15.875$, $P_2 = \$104.958$.

Consequently, when capacity utilization (Z) decreases, the cost of capital (P_1) and the operating costs (P_2) increase in accordance with an inversely proportional relationship. The growth in operating costs (P_2) occurs with a greater coefficient than for the cost of capital (P_1). The relationship between the cost of capital (P_1) and operating costs (P_2) and capacity utilization (Z) is derived from Eq. (7). The total weight of goods delivered per year ($M(t)$) is linearly related to pipeline capacity utilization (Z) and is presented in the denominator of Eq. (7). The coefficients reflecting the influence of Z on P_1 are determined based on the results obtained from Eqs. (1) and (2), while the coefficients regarding the impact of Z on P_2 are derived from Eqs. (3) and (4). When pipeline operates at full capacity ($Z = 100\%$) the total costs (W_C) of the transportation of one ton for ESPO-1 is \$ 24.167 and when pipeline operates at minimum capacity ($Z = 20\%$) is \$ 120.833. We can see that the cost increases by almost 5 times depending on the capacity utilization.

Cost components for the pumping of one ton of oil (W_C) for the ESPO-2 are shown in Fig. 4.

Fig. 4 demonstrates that the total costs $W_C = P_1 + P_2$ of the transportation of one ton of oil for ESPO-2 grow actively along with a decrease in the pipeline capacity utilization. When the pipeline capacity utilization $Z = 100\%$ (the cost of capital) $P_1 = \$3.240$ and (the operating costs) $P_2 = \$16.478$; when $Z = 80\%$, $P_1 = \$4.050$, $P_2 = \$20.598$; $Z = 60\%$, $P_1 = \$5.400$, $P_2 = \$27.464$; $Z = 40\%$, $P_1 = \$8.100$, $P_2 = \$41.196$; when $Z = 20\%$, $P_1 = \$16.200$, $P_2 = \$82.392$. Just like for ESPO-1, when capacity utilization (Z) decreases, the cost of capital (P_1) and operating costs (P_2) increase in accordance with an inversely proportional relationship. The coefficient of this relationship is greater for operating costs (P_2) than for the cost of capital (P_1). These dependences for $P_1(Z)$ and $P_2(Z)$ stem from Eq. (7). As $M(t)$ varies linearly with Z and is presented in the denominator of Eq. (7), the coefficients are determined by different relationships. The coefficient for the dependence of $P_1(Z)$ is determined based on the values obtained through Eqs. (1) and (2), while for $P_2(Z)$, it is determined using Eqs. (3) and (4). When pipeline operates at full capacity ($Z = 100\%$) the total costs of the transportation of one ton of oil for ESPO-2 is \$ 19.718, and when pipeline operates at minimum capacity ($Z = 20\%$) is \$ 98.592. There is a 5-fold increase in costs.

The next segment of transportation, after the oil is pumped through ESPO-1 and 2 and the port oil terminal, is maritime delivery. This

paper discusses the possibilities of sea transportation using three types of marine tankers: Panamax, Aframax and Suezmax. Tanker characteristics are presented in Table 1³. For maritime transport, in addition to the costs of capital ($W_1(t) + W_2(t)$) and operating expenses ($W_3(t) + W_4(t)$), a substantial cost component is the fuel costs in $W_5(t)$ Eqs. (1)–(6).

By using formulas (1)–(7), the diagram on Fig. 5 shows the tanker cost components attributed to the transportation of one ton of oil (P_1 (P1 on Fig. 5) is the cost of capital, P_2 (P2 on Fig. 5) is the operating costs, P_3 (P3 on Fig. 5) is the fuel cost for tankers).

Fig. 5 indicates the lower costs for large tankers at distribution of tanker cost components for the transportation of one ton of oil among the vessel types under review: for Panamax $\sum_{q=1}^3 P_q = \$4.442$, ($P_1 = \0.976, $P_2 = \$1.623$, and $P_3 = \$1.844$); for Aframax $\sum_{q=1}^3 P_q = \$4.101$ ($P_1 = \$0.699$, $P_2 = \$1.056$, and $P_3 = \$2.346$); and for Suezmax $\sum_{q=1}^3 P_q = \$3.424$ ($P_1 = \$0.587$, $P_2 = \$0.821$, and $P_3 = \$2.016$).

The data obtained shows that the share of the cost of capital (P_1) in the cost of transporting one ton of oil by tanker ($\sum_{q=1}^3 P_q$) for Panamax was 22 %, the operating costs (P_2) accounted for 37 %, and fuel costs (P_3) represented 41 %. For Aframax, the cost of capital (P_1), operating costs (P_2), and fuel costs (P_3) represented 17 %, 26 %, and 57 % respectively in the tanker transportation costs per ton ($\sum_{q=1}^3 P_q$). In the case of Suezmax tankers, the respective shares were 17 %, 24 %, and 59 %. Consequently, as the deadweight tonnage (dwt) of the tanker increases, the share of capital costs and operating costs decreases, while the contribution of fuel costs rises. Fuel costs (P_3) turned out to be the most significant component of tanker costs. For Panamax they were slightly less than half (41 %), for Aframax they already exceeded half (57 %), and for Suezmax tankers they accounted for 59 % of all costs. At the same time, Suezmax tankers exhibited the lowest total costs for transporting one ton of oil.

Let us compare the costs of the pipeline and marine components for the transportation of one ton of oil over a 100 km distance. Total delivery cost of one ton of oil in Equations ((1)–(7), and (10)) for 100 km for each component of a multi-modal transport ($\sum_{k=1}^3 W_C^{100km}(k)$) are shown in Fig. 6.

Each column in Fig. 6 represents the total costs of the transportation participants. The breakdown of the total cost by its main components are as follows.

At the first stage ($k = 1$), oil was supplied via ESPO-1, and when transported over 100 km, the cost per ton $W_C^{100km}(1)$ was \$0.895; at the second stage ($k = 2$), delivery took place via ESPO-2, and the cost of

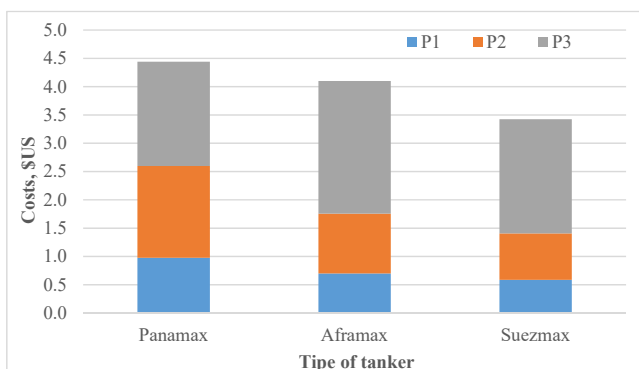


Fig. 5. Cost components for the delivery of one ton by tankers.

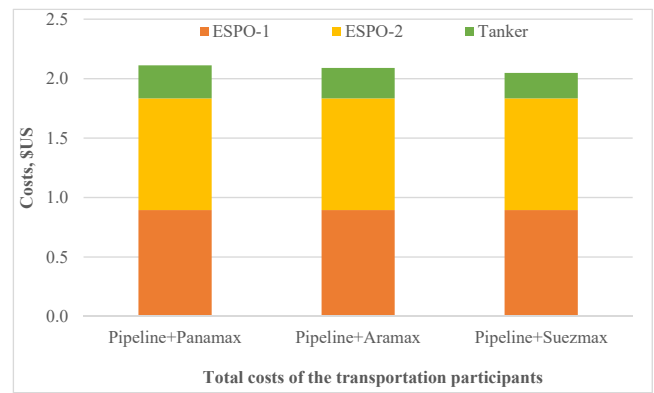


Fig. 6. Cost components for delivery of one ton per 100 km through ESPO-1,2 and tanker types.

delivering one ton per 100 km ($W_C^{100km}(2)$) was \$0.939. At the sea transportation stage ($k = 3$), when delivering one ton per 100 km, the costs $W_C^{100km}(3)$ for the tankers used were as follows: Panamax tanker - \$0.278, Aframax tanker - \$0.256, and Suezmax tanker - \$0.214.

The cost per ton for delivery via ESPO-1 is 5 % less than the cost of transporting one ton of oil via ESPO-2. The ESPO-1 pipeline facilitates the delivery of oil from Russian fields to the Chinese Skovorodino-Daqing pipeline and ESPO-2, which leads to the oil transshipment port. The cost ratio for transporting one ton of oil per 100 km via ESPO-1 compared to a Panamax tanker is 3.2 times, and via ESPO-2 is 3.4 times. For an Aframax tanker, the cost ratios are 3.5 via ESPO-1 and 3.7 via ESPO-2. For a Suezmax tanker, the ratios are 4.2 via ESPO-1 and 4.4 via ESPO-2. The results obtained show that the total cost of transporting one ton of oil for 100 km by using ESPO-1 and ESPO-2 pipelines are 6–8 times greater than the cost for similar maritime shipping. This circumstance makes the use of the maritime component more profitable for the operation of multimodal delivery routes.

5. Discussion

The calculation results presented in Figs. 1–6 show the trends in the costs of the project ESPO which we consider in this paper, using the proposed methodology. The interest rate in the range of 0 %–6 % is quite representative for the project under consideration. The cost of oil transportation through ESPO-1 increases linearly by 29 % (Fig. 1) and through ESPO-2 by 36 % (Fig. 2), with interest rates ranging from 0 % to 6 %. This increase occurs due to the growth of the cost of the capital (P_1) with the value of operating costs (P_2) unchanged.

The pipeline capacity utilization (Z) ranges from 20 % to 100 % (Figs. 3–4). At the same time, the increase in the cost of oil transportation through ESPO by $Z = 60\%$ compared to the cost of oil transportation at 100 %, is 1.7 times, while for $Z = 40\%$ it is 2.5 times, and for $Z = 20\%$ it is 5 times. Therefore, the pipeline capacity utilization is a rather sensitive parameter that significantly affects the value of the cost of oil transportation.

The cost of oil transportation varies with the carrying capacity of tankers in maritime transportation, as shown in Fig. 5. For Panamax, it is \$4.442, for Aframax is \$4.142, and for Suezmax is \$3.424. Both capital costs (P_1 : \$0.976, \$0.699, \$0.587) and operating costs (P_2 : \$1.623, \$1.056, \$0.821) decrease sequentially for Panamax, Aframax, and Suezmax as the deadweight tonnage (dwt) of the tankers in use increases. The fuel cost (P_3 : \$1.844, \$2.016) for Panamax and Suezmax is lower than the fuel cost for Aframax tankers (P_3 : \$2.346). Suezmax has the lowest total costs for transporting one ton of oil.

The data in Fig. 6 shows the ratios of the costs for transporting 1 ton per 100 km by pipeline and maritime transport in the project under consideration. The share of the cost for transporting 1 ton per 100 km

³ For more information on those types of tankers, see: <https://www.marineinsight.com/types-of-ships/panamax-and-afamax-tankers-oil-tankers-with-a-difference/>

via Panamax in the total costs of oil transportation per 100 km of the project under consideration is 13 %, for Aframax is 12 %, and for Suezmax is 10 %. For the ESPO-1 and ESPO-2 pipelines, these shares are 42 %-44 % and 45 %-46 %, respectively. Therefore, a significant portion (87 %-90 %) of the total cost for transporting 1 ton per 100 km in the ESPO project consists of the cost of pipeline transport. Therefore, the oil flows from the Siberian oil fields as well as the value of the interest rate are important to determine the total cost of multimodal delivery. The data in Fig. 6 are based on the assumption that the ESPO pipeline is 100 % loaded. In case of a decrease in pipeline load or an increase in the interest rate, a significant increase in the costs of the pipeline component for transportation can occur (Figs. 3–4). Therefore, in case of underutilization of the ESPO pipeline or an increase in the interest rate, excessive costs for pipeline transport compared to maritime transport may become even more significant.

A comparison of the data obtained (Fig. 6) for pipeline and sea delivery shows that the cost of transportation of one ton per 100 km when using a Panamax tanker is \$0.661, for an Aframax tanker is \$0.683, and for a Suezmax tanker is \$0.725 less than the cost of the ESPO-2 pipeline. The entire volume of oil (50 million tons) planned for delivery through the ESPO-2 pipeline is then transported by sea. Therefore, even at full capacity utilization, reducing the length of the pipeline route by 100 km in favor of sea delivery can result in an annual cost reduction of \$33.05 million when using a Panamax tanker, \$34.15 million when using an Aframax tanker, and \$36.25 million when using a Suezmax tanker. Furthermore, by decreasing the pipeline capacity utilization savings can increase significantly.

6. Conclusions

This paper highlighted the main cost components of the multimodal transportation of Siberian oil using pipelines and tanker fleets. Calculations show that the most significant amount of transportation costs is generated during inland transportation of oil through the pipelines.

This study fills a gap in research on cost estimation in oil supply chains. The scientific novelty of the proposed approach lies in the estimation of cost relations in hydrocarbons transportation for multimodal supply chains.

The significant cost prevalence in the pipeline transport provides significant advantages to marine transportation. Utilizing this advantage in multimodal oil delivery projects with a capacity of 50 million tons per year can bring annual savings of more than \$36 million for every 100 km of reduction in the pipeline segment of the route in favor of the offshore component. It should be taken into account when elaborating multimodal investment projects for hydrocarbons transportation.

The cost modeling approach highlighted in this paper can be used to study various options for oil supply chains. The algorithm reviewed can also be applied when developing new freight rates and estimating existing ones in multimodal oil supply projects, as well as in economics and energy courses at universities.

In the future, in order to enhance the use of this methodology, it may be necessary to take into account the risks of reduced oil supply volumes, changes in financial loan rates, and rates in the maritime market over the project's lifespan.

CRedit authorship contribution statement

Vladimir Pavlovich Klepikov: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Liubov Klepikova:** Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation. **Shawkat Hammoudeh:** Supervision, Resources, Investigation, Funding acquisition, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Data availability

No data was used for the research described in the article.

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